Datavision Using Excel

Project objective

The objective of this project is to analyze product data using Excel formulas and visualization tools. It aims to demonstrate how various Excel functions — such as SUM, AVERAGE, MIN, MAX, IF, and text functions — can be applied to organize, calculate, and interpret data efficiently.

The project also focuses on using conditional formatting and charts to highlight key insights, enabling better understanding of pricing patterns and overall data performance.

Dataset description

The product detail including product name, quantity, price and total cost. It is used to analyze sales and pricing using Excel formulas.

Column description

Column	Description
A – Product ID	Unique identification code for each product (includes date and country code).
B – Product Name	Name of the product listed for analysis.
C – Brand Name	The brand or manufacturer of the product.
D – Price ($)	Selling price of each product in dollars.
E – Quantity	Number of units sold or available.
F – Category	Product classification (e.g., Electronics, Fashion, Kitchen, Outdoor).
G – Price Range	Indicates whether the product is High Price or Standard Price based on its value.
H – Day	Extracted from Product ID to represent the day of sale or record entry.
I – Country Code	Country abbreviation showing where the product is sold (e.g., US, UK, IN).
J – Month	Extracted from Product ID to represent the month of sale or record entry.

Data Analysis Formulas in Excel

Basic Data Exploration

1) SUM, COUNTA, AVERAGE

a). SUM (Total Price of the products)

- ✓ ***Formula =SUM(D2:D35)***
- ✓ ***Result: $10100.***

b). COUNTA (How many products exist in dataset)

- ✓ ***Formula =COUNTA(B2:B35)***
- ✓ ***Result =34***

C). AVERAGE (The average price of the product)

- ✓ ***Formula =AVERAGE(D2:D35)***
- ✓ ***Result =$ 297.06***

2) Minimum and Maximum Values

A).MIN (Identifying the minimum price among all products)

- ✓ ***Formula=MIN(D2:D35)***
- ✓ ***Result=$30.***

B). MAX (Identifying the maximum price among all products)

- ✓ ***Formula = Max (D2:D35)***
- ✓ ***Result = $1000.***

3). Logical Function (Created a new column name Price Range-G)

- ✓ ***Formula =IF(D2>=500,"High Price",IF(D2<500,"Standard Price"))***
- ✓ ***Result= High Price***

4) Conditional Functions – SUMIF and COUNTIF

A) SUMIF(Using SUMIF function to calculate the total price of products in the

Electronics category)

- ✓ ***Formula= Formula=SUMIF(F2:F35,"Electronics",D2:D35)***
- ✓ ***Result= $8050.***

B) COUNTIF (Using COUNTIF function to determine the number of products with a price less than $100.)

- ✓ ***Formula=***Formula=COUNTIF(D2:D35,"<100”)
- ✓ ***Result= 11***

5). Text Formatting – LEFT, RIGHT, MID

A) LEFT Function(Created a new column named Day with the first 2 characters of each 'Product ID' using the LEFT function.)

- ✓ ***Formula=LEFT(A2,2)***
- ✓ ***Result = 2***

B) RIGHT Function (Created a new column named Country Code by extracting the last 2 characters from the 'Product ID' column using the RIGHT function.)

- ✓ ***Formula =RIGHT(A2,2)***
- ✓ ***Result = US***

C) MID Function (Created a new column named Month by extracting the 4th to 6th characters from the 'Product ID' column using the MID function.)

- ✓ ***Formula=MID(A2,4,3)***
- ✓ ***Result = JAN***

Screenshot of Workbook

- → ***In G column Red colour indicates High Price and Green colour Indicates Standard Price.***
- → ***Column Created -***
- ❑ ***G –Price Range***
- ❑ ***H –Day***
- ❑ ***I – Country Code***
- ❑ ***J- Month***

	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16	16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
22	14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
26	29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

Question No: 1

Sum,Count,Average

Calculating the **total price of the products** in the dataset.
Formula =SUM(D2:D35)
$10,100

Determining **how many products exist in the dataset.**
Formula =COUNTA(B2:B35)
34

Calculating the **average price of the product.**
Formula =AVERAGE(D2:D35)
$297.06

Question No : 2

Maximum and Minimum Values

The formula that finds the **minimum price** among all products.
Formula=MIN(D2:D35)
$30

The formula that finds the **maximum price** among all products.
Formula=MAX(D2:D35)
$1,000

Question No : 3

Logical Function-IF
Column G is Price Range -Created
Formula=IF(D2>=500,"High Price",IF(D2<500,"Standard Price"))
High Price
Standard Price

Question No: 4

Conditional Functions-SUMIF and COUNTIF
Total Price of products in the Electronisc category.
Formula=SUMIF(F2:F35,"Electronics",D2:D35)
$8,050.00

Using COUNTIF function to determine the **number of product with a price less than $100.**
Formula=COUNTIF(D2:D35,"<100")
11

Question No :5

Text Formatting - Left,Right,Mid

DAY Column -Left Function Formula=LEFT(A2,2)
28

COUNTRY CODE Column -Right Function Formula =RIGHT(A2,2)
US

MONTH Column -Mid Function Formula=MID(A2,4,3)
JAN

Conclusion

The analysis helped identify top selling products and pricing trends. Excel formulas simplified data summarization and classification. This project demonstrates how Microsoft Excel can be effectively used for data analysis and visualization. By applying formulas such as SUM, AVERAGE, MIN, MAX, and IF, along with conditional formatting and logical functions, the dataset was analyzed to identify pricing patterns and category-wise performance. The project highlights the importance of Excel as a powerful tool for organizing, calculating, and interpreting data efficiently. Overall, it enhances understanding of data-driven decision-making and strengthens practical skills in spreadsheet management.

★ CREATED BY

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